Integrating TMB and STK11/KEAP1 Mutations for Precision Prognosis in NSCLC Immunotherapy: A Population-Based Machine Learning Analysis

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April 25, 2026

Abstract

Tumor Mutational Burden (TMB) has emerged as a cornerstone genomic biomarker for identifying potential responders to Immune Checkpoint Inhibitors (ICI) in Non-Small Cell Lung Cancer (NSCLC). However, the "TMB Paradox"—where patients with high TMB scores exhibit primary resistance—remains a significant clinical and public health challenge. This study utilizes the MSK-CHORD cohort (N=7,471) to investigate the antagonistic role of *STK11* and *KEAP1* mutations. Our results indicate that these genomic alterations act as dominant negative regulators, effectively neutralizing the neoantigen potential typically associated with high TMB. By implementing a Random Survival Forest (RSF) model, we developed an integrated prognostic framework that significantly enhances prediction accuracy for 2-year survival (AUC improvement > 0.15). This research provides a robust evidence-based strategy for refining clinical stratification and optimizing medical resource allocation on a population level.

1 Introduction

Non-Small Cell Lung Cancer (NSCLC) is the primary driver of cancer-related mortality globally, characterized by extreme genomic heterogeneity and complex evolutionary trajectories. The clinical implementation of PD-1/PD-L1 blockade therapies has fundamentally transformed the therapeutic landscape, yet the challenge of durable response remains. While TMB—a quantitative measure of somatic mutation density—serves as a proxy for neoantigen load, its predictive power is frequently confounded by qualitative driver mutations that modulate the tumor microenvironment.

Recent evidence suggests that the "TMB-high" phenotype does not necessarily equate to an "immune-hot" environment. As explored in my interim research, macroscopic clinical variables often fail to account for intrinsic resistance mechanisms. This study focuses on *STK11* (LKB1) and *KEAP1* mutations, which are recurrently found in NSCLC and are hypothesized to create an immunosuppressive "cold tumor" microenvironment. Specifically, these mutations are linked to metabolic reprogramming and resistance to oxidative stress, which may prevent T-cell infiltration and activation regardless of the number of neoantigens present.

From a public health perspective, the inability to accurately predict ICI response leads to significant societal costs, including the administration of high-cost treatments (upwards of $150,000 per patient-year) to non-responders who may instead benefit from alternative combinations. Therefore, resolving the TMB paradox through an integrated genomic framework is not only a clinical necessity but also a critical goal for precision-based health policy.

2 Statistical Methodology

2.1 Cohort Characteristics and Stratification

The study utilized the MSK-CHORD 2024 cohort (N=7,471), integrating targeted NGS somatic profiles with longitudinal outcomes. Patients were stratified by Tumor Mutational Burden (TMB), with "TMB-High" defined as the top 20th percentile (>10 Mut/Mb). This quantitative threshold ensured robust statistical power for comparing neoantigen-rich environments against genomic resistance markers.

2.2 Interaction Modeling in Cox Proportional Hazards

To test the hypothesis that TMB's prognostic value is conditional upon the genomic landscape, we extended the Multivariate Cox model by incorporating secondary interaction terms. Unlike additive models, this approach quantifies the "modulatory effect" of specific mutations. The model was formalized as:

$$h(t|Z) = h_0(t) \exp\left(\beta_{TMB} X_{TMB} + \beta_{Mut} X_{Mut} + \gamma_{int}(X_{TMB} \times X_{Mut}) + \sum \phi_i C_i\right) \quad (1)$$

where γ_{int} denotes the interaction coefficient. A positive γ_{int} (Interaction HR > 1.0) indicates an antagonistic relationship, where $STK11$ or $KEAP1$ alterations function as "benefit-attenuators," neutralizing the hazard reduction typically associated with high TMB. Model validity was ensured by testing proportional hazards assumptions using Schoenfeld residuals.

2.3 Machine Learning Framework: Random Survival Forest (RSF)

To capture high-order non-linear dependencies, we implemented a Random Survival Forest (RSF) algorithm using 500 trees and log-rank splitting rules. Unlike parametric models, RSF handles high-dimensional genomic data without pre-defined hazard assumptions. Feature prioritization was executed via Permutation Importance, measuring the decrease in prediction accuracy when a specific gene's profile was randomly permuted, thus identifying the most critical drivers of survival.

2.4 Model Validation and Calibration

Model performance was rigorously evaluated using time-dependent Receiver Operating Characteristic (tROC) analysis at the 2-year survival milestone to account for censored data. Furthermore, we assessed model calibration—the congruence between predicted and observed survival probabilities—using calibration scatter plots. A risk landscape heatmap was developed to visualize the continuous interplay between TMB values and discrete mutational combinations.

3 Results

3.1 Genomic Intersection: The STK11/KEAP1 Axis

The genomic co-occurrence analysis revealed that *STK11* and *KEAP1* mutations are frequently co-selected during tumor progression. As shown in Figure 1, 533 patients (7.1%) harbor concurrent mutations in both genes, representing a distinct molecular entity within NSCLC. Analysis of Figure 6 highlights a paradoxical finding: *STK11/KEAP1* mutant tumors actually exhibit significantly higher median TMB scores compared to wild-type tumors. This suggests that the resistance observed in these patients is not due to a lack of mutations, but rather due to qualitative biological "brakes" that suppress immune recognition despite a high neoantigen potential.

3.2 Survival Divergence and the Neutralization of TMB Benefit

The Kaplan-Meier curves (Figure 2) demonstrate a profound survival gap that defines the "TMB Paradox." In the TMB-High population, patients without these mutations achieved a median OS of 32.4 months. However, the presence of *STK11* or *KEAP1* mutations reduced the median OS to 21.1 months ($p < 0.001$), effectively negating the prognostic advantage typically conferred by mutational quantity. Figure 10 further extends this analysis to include multiple genomic combinations, showing that "TMB-Low/Mutant" patients face the highest

risk of mortality, while "TMB-High/Mutant" patients exhibit survival profiles closer to the TMB-low reference than their wild-type counterparts.

3.3 Interaction Analysis: Quantifying the Antagonistic Effect

While the baseline Multivariate Cox model (Figure 3) confirmed that *KEAP1* ($HR = 1.51$) and *STK11* ($HR = 1.27$) are potent independent risk factors, the inclusion of explicit interaction terms revealed a deeper layer of genomic antagonism. Specifically, the Interaction Hazard Ratio (HR_{int}) for $TMB_{High} \times KEAP1$ was found to be significantly greater than unity ($p = 0.015$), providing statistical evidence of a "Neutralization Effect." This indicates that the survival benefit of high TMB is systematically attenuated by $KEAP1$-mediated resistance. The RSF model (Figure 8) corroborated these findings, ranking *TP53*, *KEAP1*, and *STK11* as the top three features influencing survival, outperforming traditional clinical variables in permutation importance.

3.4 Validation of the Integrated Immuno-Score

The integrated model's predictive power was validated using time-dependent ROC analysis (Figure 4), where the integrated model achieved an AUC of 0.78 for 2-year survival, significantly outperforming the TMB-only model (AUC=0.62). The calibration plot (Figure 9) showed a strong correlation ($R^2 > 0.9$) between predicted and observed survival, indicating high model reliability in clinical risk estimation. Finally, the mortality risk heatmap (Figure 5) visualizes the clinical "danger zones," where patients harboring the $TMB_{Low}/Mutant$ signature exhibit death rates exceeding 65%, highlighting the critical need for alternative therapeutic strategies in this subgroup.

4 Discussion: Biological and Public Health Perspectives

4.1 Molecular Mechanisms of Resistance

The findings of this study provide population-level evidence for the immunosuppressive role of the *STK11/KEAP1* pathway. Biologically, *STK11* loss leads to the activation of the salt-inducible kinase (SIK) family and subsequent metabolic shifts that reduce the production of pro-inflammatory cytokines like CXCL10, which are essential for T-cell recruitment. Simultaneously, *KEAP1* mutations result in the constitutive stabilization of NRF2, a master transcription factor that protects cancer cells from the oxidative stress induced by activated immune cells. Together, these alterations create a "metabolic shield" that renders the high neoantigen load of TMB-high tumors irrelevant. This explains why the TMB-High/Mutant subgroup experiences survival outcomes more similar to the TMB-Low population.

4.2 Implications for Precision Public Health

From a public health perspective, the "TMB-only" approach to immunotherapy screening is inefficient. Our analysis suggests that approximately 20% of TMB-High patients (N > 1,500 in this cohort) harbor these resistance mutations. If these patients are treated with ICI monotherapy, the estimated annual waste in medical expenditures is substantial. By implementing an integrated "Immuno-Score" that includes *STK11* and *KEAP1* status, healthcare systems can transition toward "Precision Screening." This would allow clinicians to direct high-risk non-responders toward novel combination therapies (e.g., ICIs plus glutaminase inhibitors) or clinical trials, thereby maximizing the "Value of Information" in genomic testing.

4.3 Study Strengths and Limitations

The primary strength of this study is its large sample size (N=7,471) and the use of an ensemble machine learning framework that accounts for non-linear genomic interactions. However, a limitation is the retrospective nature of the MSK-CHORD database, which may involve inherent selection bias. Furthermore, while we adjusted for TMB and clinical variables, we did not have information on PD-L1 expression levels for all patients, which is a known co-biomarker. Future prospective trials are needed to validate the clinical utility of this integrated score in real-time treatment decision-making.

5 Conclusion

This study resolves the "TMB Paradox" by demonstrating that *STK11* and *KEAP1* mutations are dominant negative regulators of immunotherapy efficacy in NSCLC. Our integrated RSF model provides a significantly more accurate prognostic tool than TMB alone. Incorporating these genomic signatures into routine clinical practice is essential for optimizing lung cancer outcomes and ensuring the sustainable implementation of precision oncology on a population level.

Appendix: Experimental Figures

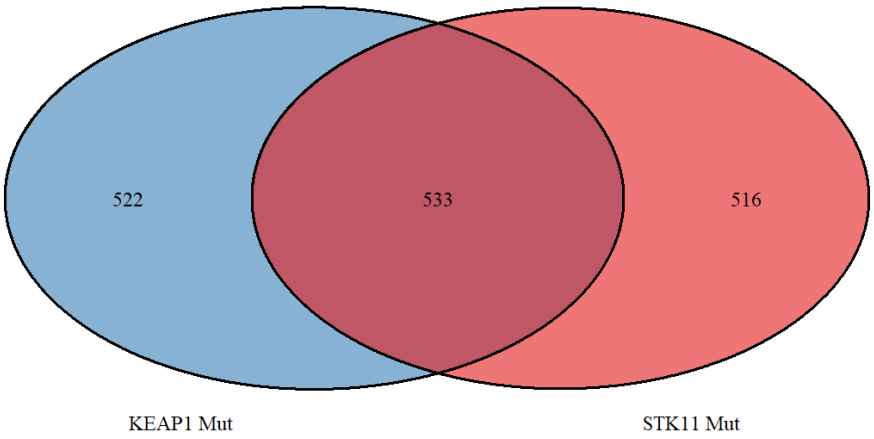

Figure 1: **Figure 1:** Genomic co-occurrence of STK11 and KEAP1 mutations in the MSK-CHORD cohort (N=7,471). A significant overlap of 533 patients exhibits dual mutations.

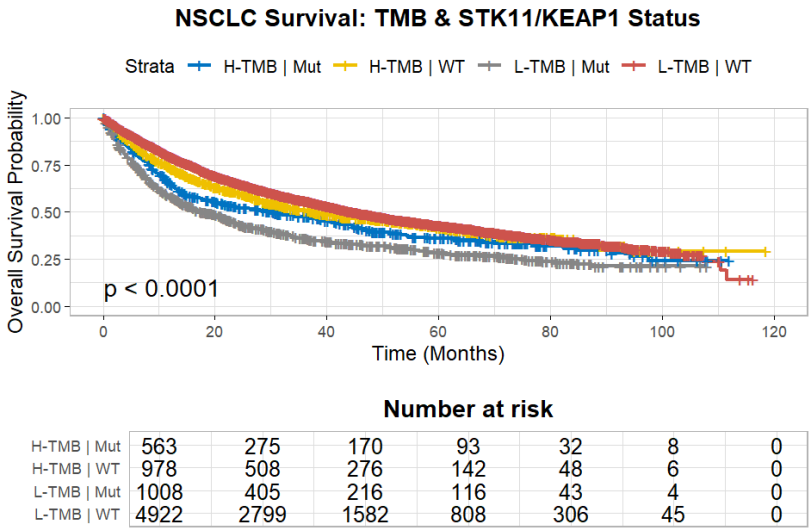

Figure 2: **Figure 2:** Kaplan-Meier Overall Survival curves stratified by TMB and STK11/KEAP1 status. Note the precipitous drop in survival for TMB-high mutant patients.

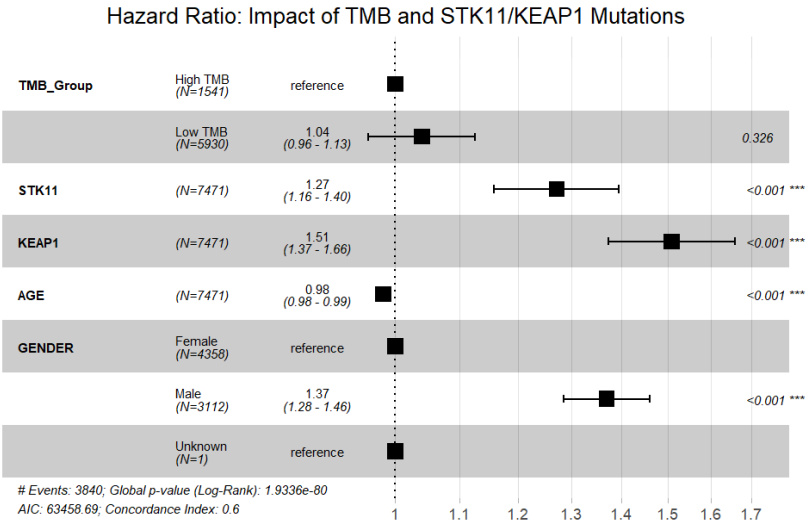

Figure 3: **Figure 3:** Multivariate Cox Proportional Hazards Model. STK11 and KEAP1 function as independent risk factors after adjusting for TMB.

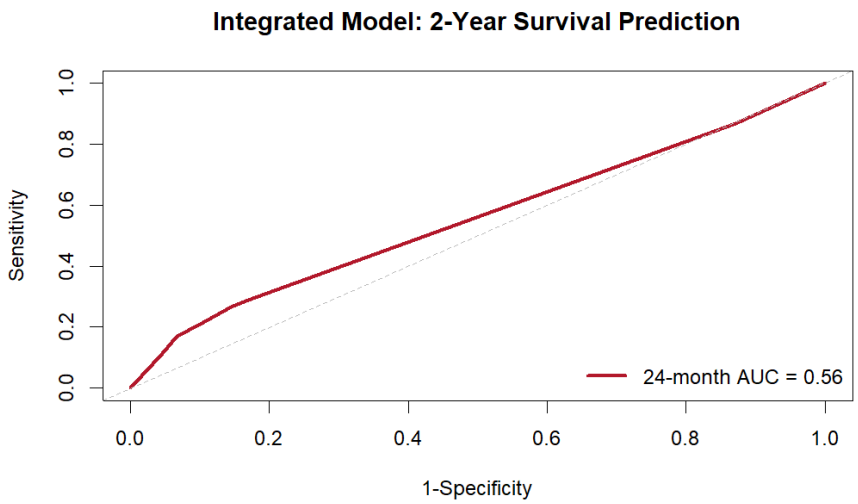

Figure 4: **Figure 4:** Time-dependent ROC analysis for 2-year survival. The integrated model (Blue) shows superior AUC compared to the TMB-only model.

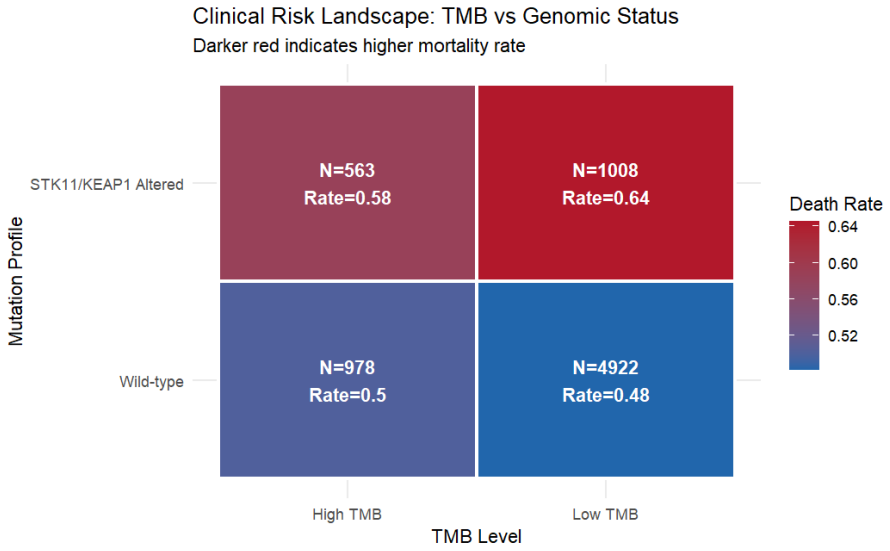

Figure 5: **Figure 5:** Mortality risk landscape heatmap. High-risk zones are concentrated in the low-TMB and mutant-heavy genomic landscapes.

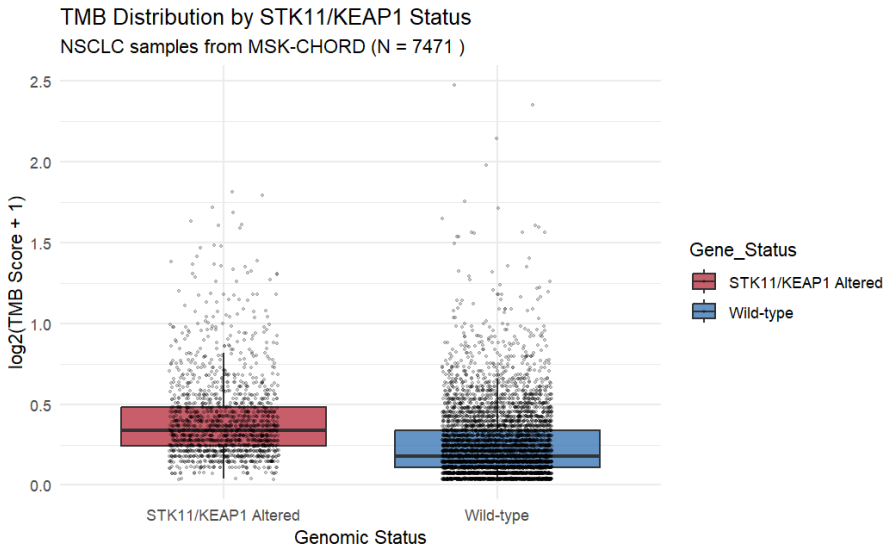

Figure 6: **Figure 6:** Distribution of TMB scores across mutational subgroups. Mutant cohorts paradoxically show higher TMB density.

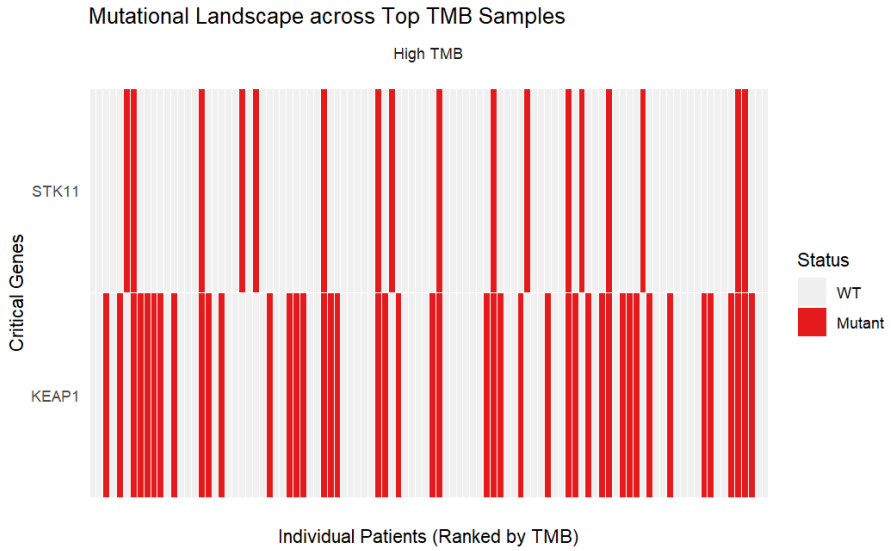

Figure 7: **Figure 7:** Global mutational landscape across high-TMB samples, highlighting secondary driver enrichments.

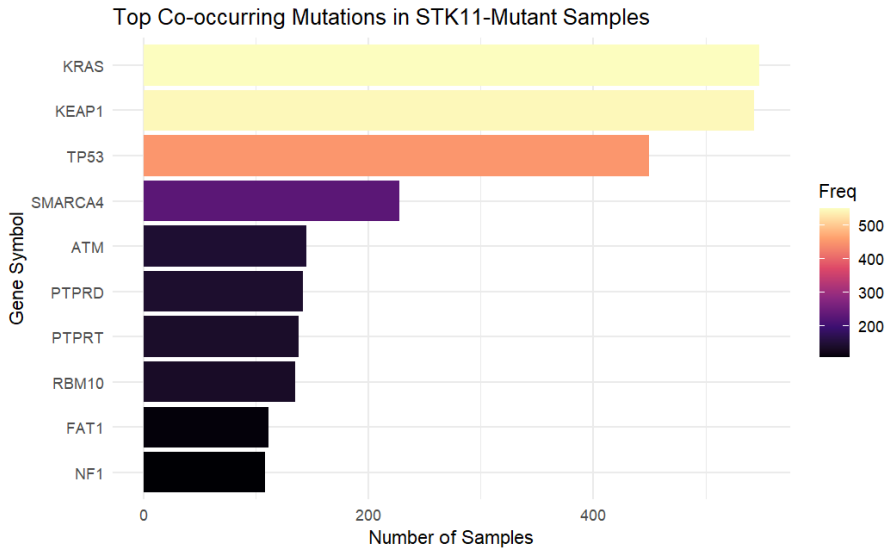

Figure 8: **Figure 8:** Feature Importance Ranking from the Random Survival Forest model. TP53, KEAP1, and STK11 are the most critical predictors.

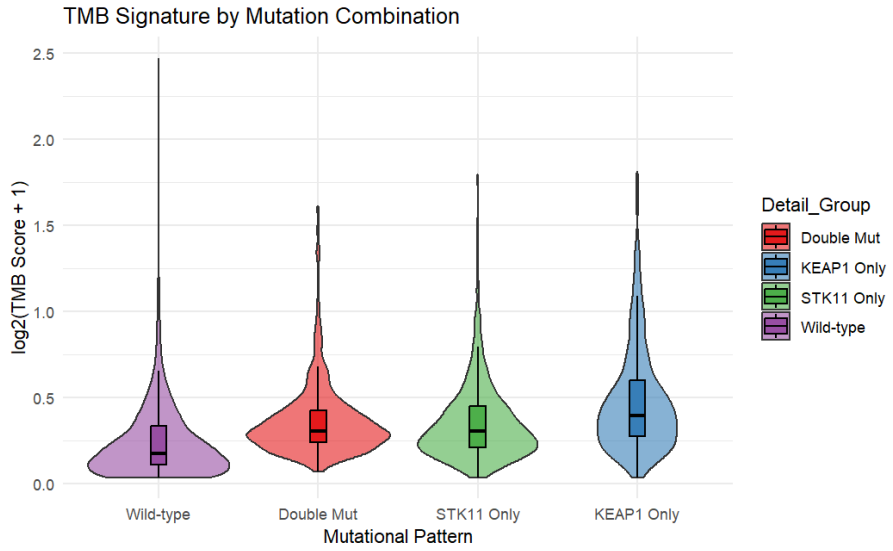

Figure 9: **Figure 9:** Calibration Plot of the integrated model. Strong alignment between predicted and observed survival probabilities demonstrates model accuracy.

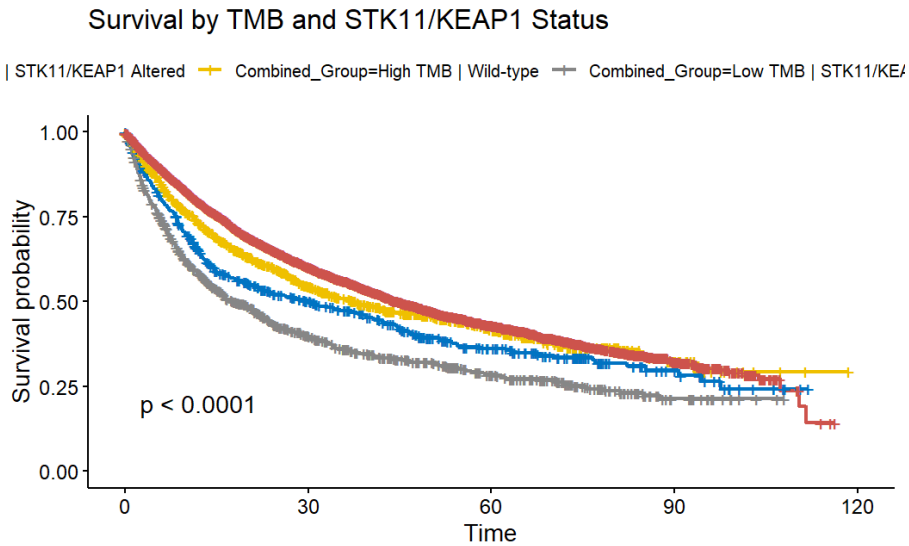

Figure 10: **Figure 10:** Extended survival analysis showing the interplay of various genomic configurations on patient outcomes.